

Binary Carry Structure in Iterated Lacunary Series

A complete modulo-four theorem for OEIS A168362,
an exact exception criterion, and a logarithmic counting law

Research note prepared for Vladimir Reshetnikov with OpenAI

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Abstract

Let

$$F(x) = \sum_{j \geq 0} x^{2^j} = x + x^2 + x^4 + x^8 + \cdots, \quad G_m(x) = F^{\circ m}(x),$$

and let OEIS A168362 be

$$a(n) = [x^n] G_n(x).$$

The OEIS entry records two 2026 conjectural observations: the ratio $a(n+1)/a(n) \sim en$, and a sparse pattern for $a(n) \bmod 4$. This paper settles the second observation completely and strengthens it from the diagonal to the entire iteration array.

The main structural result is that

$$[x^N] G_m(x) \equiv 0 \pmod{4} \quad \text{whenever } \text{wt}_2(N) \geq 3,$$

for every $m, N \geq 1$, where $\text{wt}_2(N)$ is the number of 1's in the binary expansion of N . The coefficients at powers of two and at sums of two powers of two are then described by exact finite bit formulas. Specializing to the diagonal gives a complete classification. Besides the four initial exceptions 2, 3, 4, 6, write $n = 2^r + 2^s$ with $r > s \geq 1$ and $d = r - s$. For $r \geq 3$, one has $a(n) \equiv 2 \pmod{4}$ precisely when $d \bmod 2^s$ belongs to an explicitly computable residue set R_s . The first exceptions are

$$2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, 1028, 1032, 2056, 2064, \dots$$

Thus the theorem confirms every exception printed in the OEIS entry and predicts all subsequent ones.

Finally, if $E(X)$ counts the exceptional indices up to X , then

$$E(X) = \kappa \log_2 X + O((\log \log X)^3), \quad \kappa = \sum_{s \geq 1} \frac{|R_s|}{2^s} = 2.323930059308106 \dots$$

All finite calculations in the article are reproduced by a standard-library Python verifier included with the source archive. The argument itself is symbolic and does not depend on the computation.

Status and proof boundary. This is an AI-assisted, unrefereed research manuscript. The congruence statements are proved in the text. The accompanying program independently checks direct truncated composition modulo 4 through degree and iterate 256, checks the published initial terms, and regenerates all residue tables. No claim of formal verification in Lean or of publication priority is made. Independent mathematical review is recommended before an OEIS submission.

[Write note, batch 73O1: filed in ProveIt's research-report collection; provenance and status in Section 1.3.]

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1 The OEIS problem and the new results

The sequence A168362 is defined by the main diagonal of the iteration table of $F(x) = \sum_{j \geq 0} x^{2^j}$:

$$a(n) = [x^n] F^{\circ n}(x). \quad (1)$$

Its initial values are

1, 2, 6, 34, 280, 3010, 39984, 634040, 11704548,

The current OEIS entry records the conjectural ratio

$$\frac{a(n+1)}{a(n)} \sim en \quad (2)$$

and the observed congruence pattern

$$a(n) \equiv 2 \pmod{4} \quad \text{at } n = 2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, \dots, \quad (3)$$

with apparent divisibility by 4 elsewhere; see [1].

The purpose of this article is arithmetic rather than asymptotic. The ratio problem (2) remains open here. The modulo-four observation, however, is completely resolved. The result is stronger than a list of indices: it gives a short bitwise decision procedure, describes every entry of the two-dimensional iteration table modulo 4, and yields an asymptotic law for the number of exceptions.

1.1 Statement of the diagonal classification

For nonnegative integers u, v , $u \& v$ and $u | v$ denote bitwise AND and OR. Let $\text{wt}_2(u)$ be the binary Hamming weight. For $s \geq 1$, put $M_s = 2^s - 1$ and define a subset $R_s \subseteq \mathbb{Z}/2^s\mathbb{Z}$ by

$$\mathbf{1}_{d \in R_s} \equiv \mathbf{1}_{d \equiv 0 \pmod{2^s}} + \sum_{i=0}^s \mathbf{1}_{i | ((i+d) \bmod 2^s) = M_s} \pmod{2}. \quad (4)$$

This is an exact finite definition involving only $s + 2$ Boolean tests.

Theorem 1.1 (Complete modulo-four classification for A168362). *For $n > 1$, $a(n)$ is even. Moreover, $a(n) \equiv 2 \pmod{4}$ if and only if one of the following holds:*

- (i) $n \in \{2, 3, 4, 6\}$;
- (ii) $n = 2^r + 2^s$ with $r > s \geq 1$, $r \geq 3$, and $r - s \pmod{2^s} \in R_s$.

Every other $n > 1$ satisfies $a(n) \equiv 0 \pmod{4}$.

The residue sets begin

$$\begin{aligned} R_1 &= \emptyset, \\ R_2 &= \{0, 2\}, \\ R_3 &= \{0, 1, 2, 4, 6, 7\}, \\ R_4 &= \{0, 7, 9, 10, 11, 12, 14, 15\}, \\ R_5 &= \{0, 21, 22, 23, 27, 28, 30, 31\}, \\ R_6 &= \{0, 51, 54, 57, 59, 60, 62, 63\}. \end{aligned}$$

For example, $s = 2$ gives the particularly simple family

$$a(2^r + 4) \equiv 2 \pmod{4} \iff r \text{ is even, } r \geq 3.$$

This produces 20, 68, 260, 1028, 4100, 16388, ...

[Write note, batch 7301.] Since $R_1 = \emptyset$, case (ii) of Theorem 1.1 never holds with $s = 1$: the exceptional two-bit indices all have $s \geq 2$. The condition is stated as $r > s \geq 1$ in the source and is kept.

1.2 A theorem for the whole iteration array

The diagonal theorem follows from a more general statement. Write

$$g_m(N) = [x^N] G_m(x), \quad G_m(x) = F^{\circ m}(x). \quad (5)$$

Theorem 1.2 (Binary-weight support modulo four). *For every $m, N \geq 1$,*

$$\text{wt}_2(N) \geq 3 \implies g_m(N) \equiv 0 \pmod{4}.$$

In addition, $g_m(N)$ is odd only when N is a power of two. Exact formulas for the remaining one-bit and two-bit exponents are given in Theorem 5.2.

This support theorem is not a numerical coincidence. It is a consequence of the fact that, modulo 4, the only cross terms that survive a 2^k -th power are balanced products of two monomials. That two-adic fact is the core of the proof.

1.3 Place in the ProveIt collection

[Write note, batch 73O1, 1 October 2026.] This report is filed in ProveIt’s research-report collection, in the directory

SetTheory/Cardinals/docs/reports/congruences-and-valuations/
iterated-series/a168362-lacunary-iterates-mod4.

Its README there is the guide to the files.

Provenance. The report is built from one manuscript: manuscript 49 of batch 73 (cluster O1) of ProveIt’s incoming-reports intake, the archive `ProveIt_A168362_Mod4_Theorem.zip`, which arrived in commit `3a9518c52` and was placed by commit `9df4ba51a`. The manuscript names no pinned commit; it says the repository [10] was inspected on 1 October 2026. The text is printed as delivered. The write changed only the label names, which now carry the prefix `lim:`, and added the notes marked [Write note]. Nothing was merged into it; no other manuscript of the batch treats this sequence. The delivered bibliography also lists two general references, on formal power series [8] and on analytic combinatorics [9], which the delivered text never cites; they are kept as background.

Status. The manuscript was prepared with OpenAI. It is unrefereed, and nothing in it is formalized in Lean or any other proof assistant. The intake reran the shipped verifier on a copy (exit code 0; the three data files it writes equal the shipped ones, up to line endings). Independently of the delivered code, it composed F modulo 4 to degree and iterate 1100: the diagonal exceptions are 2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, 1028, 1032, as Theorem 1.1 predicts, and Theorem 1.2 holds for $m \leq 300$, $N \leq 1100$. It also recomputed $a(1), \dots, a(10)$, the sets $R_1, \dots, R_8$, the tail bound of the series for κ , and the row $n = 20$ of Table 2. It did not re-derive every proof. The OEIS statements quoted here (the comments of 22 May 2026 on A168362, and the entries A112317, A122888 and A086753) were confirmed on the live entries on 1 October 2026.

Neighbouring reports. In the same collection:

- the two other reports in `iterated-series` prove congruences for compositional iterates of other series:
`a396807-self-iterating-series` and
`compositional-tree-series-congruences`;
- `a000139-binary-carries-parity` uses the same binary-carry (Kummer–Lucas) technique for a different sequence.

There is no overlap, and no statement here depends on them.

2 Formal-series setup and binary lifts

All power series below have zero constant term, so formal composition is well-defined. We work coefficientwise; therefore every assertion remains valid after truncation to any finite degree.

For a binary series $P = \sum p_n x^n \in \mathbb{F}_2[[x]]$, let

$$[P] = \sum \tilde{p}_n x^n \in \mathbb{Z}/4\mathbb{Z}[[x]]$$

denote the canonical lift, where $\tilde{p}_n \in \{0, 1\}$. Every series $H \in \mathbb{Z}/4\mathbb{Z}[[x]]$ has a unique binary decomposition

$$H = [P] + 2[Q], \quad P, Q \in \mathbb{F}_2[[x]]. \quad (6)$$

The first component records coefficient parity, while the second records the next binary digit.

Let T be Frobenius on $\mathbb{F}_2[[x]]$:

$$T(P) = P^2.$$

Because the characteristic is two,

$$F(P) = P + P^2 + P^4 + \cdots = (1 + T)^{-1}P.$$

Here the inverse is the coefficientwise convergent geometric series $1 + T + T^2 + \cdots$.

3 The modulo-four polarization identity

3.1 A multinomial valuation lemma

Lemma 3.1 (Only balanced pairs survive). *Let $z_1, z_2, \dots$ be commuting indeterminates of positive degree. For $k \geq 1$,*

$$\left(\sum_j z_j \right)^{2^k} \equiv \sum_j z_j^{2^k} + 2 \sum_{i < j} (z_i z_j)^{2^{k-1}} \pmod{4}. \quad (7)$$

Proof. Consider a multinomial term with exponents $e_1, e_2, \dots$ summing to 2^k . Legendre's formula gives

$$v_2 \left(\frac{(2^k)!}{\prod_j e_j!} \right) = \sum_j s_2(e_j) - 1,$$

where s_2 is binary digit sum. The valuation is zero exactly when one $e_j = 2^k$ and all other exponents vanish. The valuation is one exactly when the total binary digit sum of the nonzero exponents is two. Since their sum is a single power of two, this forces exactly two nonzero exponents, both equal to 2^{k-1} . Every other multinomial coefficient is divisible by 4. The balanced binomial coefficient has valuation one, hence is congruent to 2 modulo 4. This proves (7). $\square$

3.2 Polarizing the lacunary operator

Summing Theorem 3.1 over k gives a compact identity.

Corollary 3.2 (Lacunary polarization). *For positive-degree monomials z_j ,*

$$F \left(\sum_j z_j \right) \equiv \sum_j F(z_j) + 2 \sum_{i < j} F(z_i z_j) \pmod{4}. \quad (8)$$

A second elementary fact explains how the $2[Q]$ component propagates.

Lemma 3.3 (The high binary digit is carried unchanged). *For $P, Q \in \mathbb{F}_2[[x]]$,*

$$F([P] + 2[Q]) \equiv F([P]) + 2[Q] \pmod{4}. \quad (9)$$

Proof. The term with exponent one is $[P] + 2[Q]$. For every $k \geq 1$, $([P] + 2[Q])^{2^k} \equiv [P]^{2^k} \pmod{4}$, because every term containing $2[Q]$ has a factor divisible by 4. $\square$

4 The carry transform

The parity iterates are supported only on powers of two. It is therefore useful to encode a subset $S \subseteq \mathbb{Z}_{\geq 0}$ by

$$P_S(x) = \sum_{j \in S} x^{2^j} \in \mathbb{F}_2[[x]].$$

For $r \geq 0$, define the prefix count

$$N_S(r) = |S \cap \{0, 1, \dots, r\}|.$$

Definition 4.1 (Carry transform). The binary series $\mathcal{C}(P_S)$ is defined by

$$[x^{2^r}] \mathcal{C}(P_S) = \binom{N_S(r)}{2} \pmod{2}, \quad (10)$$

$$[x^{2^r+2^s}] \mathcal{C}(P_S) = \sum_{i=0}^s \mathbf{1}_{i \in S} \mathbf{1}_{i+r-s \in S} \pmod{2}, \quad r > s, \quad (11)$$

and all coefficients at exponents of binary weight at least three are zero.

Proposition 4.2 (Exact lift of one lacunary step). *For every S ,*

$$F([P_S]) \equiv [F(P_S)] + 2[\mathcal{C}(P_S)] \pmod{4}. \quad (12)$$

Proof. By Theorem 3.2,

$$F([P_S]) \equiv \sum_{j \in S} F(x^{2^j}) + 2 \sum_{\substack{i < j \\ i, j \in S}} F(x^{2^i+2^j}) \pmod{4}.$$

At exponent 2^r , the first sum has coefficient $N_S(r)$. Its low binary digit is $N_S(r) \bmod 2$, and its next digit is

$$\left\lfloor \frac{N_S(r)}{2} \right\rfloor \equiv \binom{N_S(r)}{2} \pmod{2}.$$

This gives (10). In the second sum, the exponent $2^r + 2^s$ occurs from a pair (i, j) precisely when $(i, j) = (s - k, r - k)$ for some $0 \leq k \leq s$. Reindexing by $i = s - k$ gives (11). No exponent with three binary 1's can occur. $\square$

5 Closed form for the parity iterates

Write

$$G_m(x) \equiv [P_m(x)] + 2[Q_m(x)] \pmod{4}, \quad P_m, Q_m \in \mathbb{F}_2[[x]]. \quad (13)$$

Starting with $P_0 = x$ and $Q_0 = 0$, Theorems 3.3 and 4.2 give

$$P_{m+1} = F(P_m), \quad Q_{m+1} = Q_m + \mathcal{C}(P_m) \quad \text{in } \mathbb{F}_2[[x]]. \quad (14)$$

Proposition 5.1 (Negative-binomial parity formula). *For $m \geq 1$,*

$$P_m(x) = \sum_{j \geq 0} \binom{m+j-1}{j} x^{2^j} \quad \text{in } \mathbb{F}_2[[x]]. \quad (15)$$

Equivalently, if

$$S_m = \{j \geq 0 : j \& (m-1) = 0\},$$

then $P_m = P_{S_m}$.

Proof. Over $\mathbb{F}_2[[x]]$, $F = (1+T)^{-1}$. Therefore

$$P_m = (1+T)^{-m} x = \sum_{j \geq 0} \binom{m+j-1}{j} T^j x,$$

which is (15). Lucas's theorem, or equivalently the no-carry form of Kummer's theorem, says that $\binom{m+j-1}{j}$ is odd exactly when the addition of j and $m-1$ has no binary carry, namely when $j \& (m-1) = 0$; see [7]. $\square$

Combining the recurrence with the closed form gives the promised theorem for the complete array.

Theorem 5.2 (Exact array formula modulo four). *Let $m \geq 1$.*

- (a) *If $\text{wt}_2(N) \geq 3$, then $g_m(N) \equiv 0 \pmod{4}$.*
- (b) *If N is not a power of two, then $g_m(N)$ is even.*
- (c) *For $N = 2^r$,*

$$g_m(2^r) \equiv \binom{m+r-1}{r} + 2 \sum_{\ell=1}^{m-1} \binom{\#\{0 \leq j \leq r : j \& (\ell-1) = 0\}}{2} \pmod{4}. \quad (16)$$

- (d) *For $N = 2^r + 2^s$ with $r > s \geq 0$, put $d = r - s$ and*

$$E(M, B) = \#\{0 \leq t \leq M : t \& B = 0\} \pmod{2}, \quad E(M, B) = 0 \text{ for } M < 0. \quad (17)$$

Then

$$g_m(2^r + 2^s) \equiv 2 \sum_{i=0}^s E(m-2, i \mid (i+d)) \pmod{4}. \quad (18)$$

Proof. By (14), $Q_m = \sum_{\ell=0}^{m-1} \mathcal{C}(P_\ell)$; the $\ell = 0$ carry is zero. Every P_ℓ is supported at one-bit exponents, and every $\mathcal{C}(P_\ell)$ is supported at exponents of weight at most two. This proves (a) and (b). Formula (16) follows from (10) and Theorem 5.1.

For a two-bit exponent, substitute S_ℓ into (11). With $t = \ell - 1$, membership of both i and $i + d$ in S_ℓ is equivalent to

$$t \& i = 0, \quad t \& (i + d) = 0,$$

and hence to $t \& (i \mid (i + d)) = 0$. Summing $0 \leq t \leq m - 2$ gives (18). $\square$

Remark 5.3 (A logarithmic-time evaluation of E). The parity in (17) can be evaluated without enumeration. Every admissible $t \leq M$ is either the tight word $t = M$ or has a highest bit j at which $M_j = 1$ and $t_j = 0$. Hence

$$E(M, B) = \mathbf{1}_{M \& B = 0} \oplus \bigoplus_{\substack{j: M_j = 1; \\ (M \gg (j+1)) \& (B \gg (j+1)) = 0; \\ B \bmod 2^j = 2^j - 1}} 1, \quad (19)$$

where $\oplus$ denotes XOR. Thus every entry at a two-bit exponent is computable in $O(s \log m)$ bit operations.

6 Specialization to the A168362 diagonal

We now set $m = N = n$.

6.1 All noninitial diagonal terms are even

Proposition 6.1 (Diagonal parity). *For every $n > 1$, $a(n)$ is even.*

Proof. If n is not a power of two, this follows from Theorem 5.2(b). If $n = 2^r$ with $r \geq 1$, the parity is

$$\binom{2^r + r - 1}{r} \pmod{2}.$$

By Lucas's theorem it would be odd only if $r \& (2^r - 1) = 0$. But $0 < r < 2^r$ and the lowest r binary positions of $2^r - 1$ are all 1, so $r \& (2^r - 1) = r \neq 0$. $\square$

6.2 Powers of two

Proposition 6.2 (The one-bit diagonal). *For $r \geq 1$,*

$$a(2^r) \equiv 2 \pmod{4} \iff r \in \{1, 2\}.$$

Thus only $n = 2$ and $n = 4$ are exceptional powers of two.

Proof. For $t = 0, 1, \dots, 2^r - 2$, put

$$N_t = \#\{0 \leq j \leq r : j \& t = 0\}.$$

By (16) and Theorem 6.1, the second binary digit of $a(2^r)$ is

$$\sum_{t=0}^{2^r-2} \binom{N_t}{2} \pmod{2}.$$

Interchange the sum over t with the choice of a pair $0 \leq i < j \leq r$:

$$\sum_{0 \leq i < j \leq r} \#\{0 \leq t \leq 2^r - 2 : t \& (i | j) = 0\} \pmod{2}. \quad (20)$$

The excluded value $t = 2^r - 1$ is never counted because $i | j > 0$. Over the complete set of r -bit values, the inner count is $2^{r-\text{wt}_2(i|j)}$, which is odd exactly when $i | j = 2^r - 1$. For $r = 1$, the pair $(0, 1)$ works; for $r = 2$, the pair $(1, 2)$ works. For $r \geq 3$, both i and j are at most $r < 2^{r-1}$, so their OR is smaller than $2^r - 1$. This proves the claim. $\square$

6.3 Two-bit exponents

For $n = 2^r + 2^s$, $r > s \geq 0$, Theorem 5.2 gives the exact criterion

$$\frac{a(n)}{2} \equiv \sum_{i=0}^s E(n-2, i | (i+r-s)) \pmod{2}. \quad (21)$$

The following elementary evaluation is what turns this into Theorem 1.1.

Lemma 6.3 (A nearly complete binary interval). *For $s \geq 1$ and $B \geq 0$,*

$$E(2^s - 2, B) = 1 \iff B \bmod 2^s \in \{0, 2^s - 1\}. \quad (22)$$

Proof. The complete range $0 \leq t \leq 2^s - 1$ contains $2^{s-\text{wt}_2(B \bmod 2^s)}$ values disjoint from B . This count is odd exactly when the low s bits of B are all 1. Passing from the complete range to $0 \leq t \leq 2^s - 2$ removes the all-ones value, and that removed value is disjoint from B exactly when the low s bits of B are all 0. $\square$

Proof of Theorem 1.1. Parity was proved in Theorem 6.1. Exponents with at least three binary 1's are divisible by 4 by Theorem 1.2; powers of two were settled in Theorem 6.2.

Suppose $n = 2^r + 2^s$, $r > s \geq 0$. If $s = 0$, then $n - 2 = 2^r - 1$ and (21) has the single mask $B = r$. The count over all r -bit values is odd only when $r = 2^r - 1$, which occurs only at $r = 1$. This gives $n = 3$. The remaining small case $(r, s) = (2, 1)$ gives $n = 6$ by direct substitution in (21).

Now assume $r \geq 3$ and $s \geq 1$, and put $d = r - s$. Split $0 \leq t \leq n - 2 = 2^r + 2^s - 2$ into

$$0 \leq t < 2^r, \quad t = 2^r + u, \quad 0 \leq u \leq 2^s - 2.$$

For $B_i = i | (i + d)$, the first block has even cardinality: both i and $i + d$ are at most r , so $B_i < 2^r - 1$ when $r \geq 3$, and the number of r -bit values disjoint from B_i is an even power of two. In the second block, bit r of B_i is zero, so the parity is $E(2^s - 2, B_i)$. By Theorem 6.3, this is one exactly when the low s bits of B_i are all zero or all one. The all-zero case occurs exactly when $i = 0$ and $d \equiv 0 \pmod{2^s}$. The all-one cases are exactly the tests in (4). Their XOR is the definition of R_s . $\square$

7 Constructing the residue sets

Definition (4) already yields an $O(s)$ membership test. For enumeration and counting, a symmetric-difference description is more convenient. For $0 \leq i \leq s$, let

$$D_{s,i} = \{(((M_s \text{ xor } i) | u) - i) \bmod 2^s : u \text{ is a bitwise submask of } i\}. \quad (23)$$

Here $M_s \text{ xor } i = M_s - i$ because M_s consists entirely of 1's.

Proposition 7.1 (Symmetric-difference construction).

$$R_s = \{0\} \triangle D_{s,0} \triangle D_{s,1} \triangle \cdots \triangle D_{s,s}, \quad (24)$$

where $\triangle$ denotes symmetric difference. Furthermore,

$$|R_s| \leq 1 + \sum_{i=0}^s 2^{\text{wt}_2(i)} \leq 1 + \frac{(s+1)(s+2)}{2}. \quad (25)$$

For $s \geq 3$, every nonzero $d \in R_s$ satisfies

$$d \geq 2^s - 1 - 2s. \quad (26)$$

Proof. Fix i . Put $y = (i + d) \bmod 2^s$. The condition $i \mid y = M_s$ forces all zero bits of i to be one in y , while the one bits of i are free. Thus $y = (M_s \text{ xor } i) \mid u$ with u a submask of i . This proves (24). There are $2^{\text{wt}_2(i)}$ such submasks. Since the least nonnegative integer with k one-bits is $2^k - 1$, one has $2^{\text{wt}_2(i)} \leq i + 1$, giving (25). Finally, every such y is at least $M_s - i$, so for $s \geq 3$, $y - i \geq M_s - 2s > 0$. This proves (26). $\square$

Table 1: The first residue sets and their sizes.

s	$ R_s $	R_s
1	0	$\emptyset$
2	2	0, 2
3	6	0, 1, 2, 4, 6, 7
4	8	0, 7, 9, 10, 11, 12, 14, 15
5	8	0, 21, 22, 23, 27, 28, 30, 31
6	8	0, 51, 54, 57, 59, 60, 62, 63
7	12	0, 113, 114, 116, 117, 119, 120, 121, 123, 124, 126, 127
8	12	0, 239, 241, 242, 244, 245, 248, 249, 251, 252, 254, 255

The lower bound (26) explains why only small s appear in the early exception list. Although R_s is periodic modulo 2^s , its nonzero residues lie in a short interval near the top of that modulus.

8 The infinite exception sequence

Applying Theorem 1.1 gives

$$\begin{aligned} &2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, 1028, 1032, 2056, 2064, \\ &4100, 4104, 8200, 8208, 16388, 16400, 32776, 32784, 65540, 65552, \\ &131080, 262148, 262152, 262160, 524296, 524304, 1048580, \dots \end{aligned}$$

The terms through 520 are exactly those displayed in the OEIS comment. The next two predictions are

$$a(1028) \equiv a(1032) \equiv 2 \pmod{4},$$

followed by 2056 and 2064.

Corollary 8.1 (A compact family description). *For every $r \geq 3$,*

$$\begin{aligned} a(2^r + 4) &\equiv 2 \pmod{4} && \iff && r \equiv 0 \pmod{2}, \\ a(2^r + 8) &\equiv 2 \pmod{4} && \iff && r - 3 \equiv 0, 1, 2, 4, 6, 7 \pmod{8}. \end{aligned}$$

The corresponding statements for $2^r + 2^s$ are periodic in $r - s$ with period 2^s .

9 How many exceptions are there?

Define

$$\mathcal{E}(X) = \#\{2 \leq n \leq X : a(n) \equiv 2 \pmod{4}\}. \quad (27)$$

The residue classification yields a convergent density constant on the logarithmic scale.

Definition 9.1. Let

$$\kappa = \sum_{s \geq 1} \frac{|R_s|}{2^s}. \quad (28)$$

The series converges absolutely by (25).

Theorem 9.2 (Logarithmic counting law). As $X \rightarrow \infty$,

$$\mathcal{E}(X) = \kappa \log_2 X + O((\log \log X)^3). \quad (29)$$

Numerically,

$$\kappa = 2.323930059308106 \dots \quad (30)$$

Proof. First take $X = 2^L - 1$. Apart from the four isolated exceptions 2, 3, 4, 6, every exceptional index has the form $n = 2^{s+d} + 2^s$ with $s \geq 2$, $d \geq 1$, $s + d \leq L - 1$, and $d \bmod 2^s \in R_s$. Hence

$$\mathcal{E}(2^L - 1) = 4 + \sum_{s=2}^{L-2} \#\{1 \leq d \leq L - 1 - s : d \bmod 2^s \in R_s\}. \quad (31)$$

For fixed s and $Y \geq 0$, periodic counting gives

$$\#\{1 \leq d \leq Y : d \bmod 2^s \in R_s\} = \frac{|R_s|}{2^s} Y + O(|R_s|). \quad (32)$$

By (26), a summand is zero once $2^s - 1 - 2s > Y$ and $2^s > Y$. Thus only $s = O(\log L)$ contribute. Summing the error in (32) and using $|R_s| = O(s^2)$ gives $O((\log L)^3)$. Replacing $Y = L - 1 - s$ by L changes the main sum by a bounded amount because $\sum s|R_s|/2^s$ converges. The tail of (28) beyond $s = 2\lceil \log_2 L \rceil$ contributes $o(1)$ after multiplication by L . This proves

$$\mathcal{E}(2^L - 1) = \kappa L + O((\log L)^3).$$

For arbitrary $X \in [2^L, 2^{L+1})$, only exponents with top bit L can be added. Again (26) restricts their lower-bit positions to $O(\log L)$ possibilities. This is absorbed by the same error term, and $L = \log_2 X + O(1)$ proves (29).

For the decimal, summing through $s = 64$ gives

$$\sum_{s=1}^{64} \frac{|R_s|}{2^s} = \frac{10717235762314325637}{4611686018427387904} = 2.323930059308106699 \dots$$

From (25), the remaining tail is at most

$$\sum_{s>64} \frac{1 + (s+1)(s+2)/2}{2^s} = \frac{64^2 + 7 \cdot 64 + 16}{2^{65}} = \frac{285}{2^{61}} < 1.24 \times 10^{-16}.$$

□

Remark 9.3. The exceptional set has natural density zero for the trivial reason that it is contained in the integers of binary weight at most two. Theorem 9.2 is sharper: among the $\asymp (\log X)^2$ integers of binary weight two below X , only $\asymp \log X$ are exceptional.

10 Independent exact verification

[Write note, batch 7301.] By default the program writes into the `data/` directory beside its own parent directory, so a run in the collection would overwrite the shipped records; its option `-root` selects another directory. The README gives a rerun on a copy. The file `data/verification_run.txt` is the captured standard output of the recorded run, which prints the same JSON record; it is byte-identical to `data/verification.json`.

The archive contains `code/verify_a168362.py`. It uses only the Python standard library and has two deliberately separate layers.

1. It directly composes truncated dense polynomials in $\mathbb{Z}/4\mathbb{Z}[x]$ by repeated squaring, without using the carry transform. Through degree and iterate 256, it checks:
 - the diagonal classification;
 - divisibility by 4 at every exponent of binary weight at least three;
 - the parity formula at powers of two;
 - the exact two-bit formula (18).
2. It independently regenerates the sets R_s from the symmetric difference (24), the first exceptions through 10^7 , and the partial sum for κ .

The direct 256×256 table has SHA-256 digest

df15e35655f2aaf2905219425b6ebcf63a2b8a8daf0144380b1e21824cf091bf.

The program also performs direct integer composition for the 21 terms currently printed on the OEIS page and obtains an exact match.

A typical run is

```
python code/verify_a168362.py --limit 256
```

and writes the machine-readable files `data/verification.json`, `data/residue_sets.csv`, and `data/exceptional_indices_to_10m.txt`.

11 What the result says about the ratio conjecture

The congruence theorem does not prove (2); the arithmetic and asymptotic questions live at different scales. Still, a few rigorous comparisons and numerical observations help delimit the next problem.

Let $B(x) = x + x^2$ and $H(x) = x/(1 - x)$. Coefficientwise,

$$B(x) \preceq F(x) \preceq H(x).$$

Composition preserves these inequalities for series with nonnegative coefficients. Since

$$H^{\circ n}(x) = \frac{x}{1 - nx},$$

we obtain

$$a(n) \leq n^{n-1}. \tag{33}$$

The lower comparator

$$b(n) = [x^n] B^{\circ n}(x)$$

is OEIS A112317 [4]. The associated array has a word interpretation: $[x^k] B^{\circ n}(x)$ counts words of length $k - 1$ on $[n]$ in which two occurrences of i are separated by a larger letter [5]. Every word of length $n - 1$ with distinct letters is admissible, and there are $n!$ such words. Therefore

$$n! \leq b(n) \leq a(n) \leq n^{n-1}. \quad (34)$$

In particular,

$$\log a(n) = n \log n + O(n).$$

Direct exact computation through $n = 300$ gives the following diagnostics. Here

$$R_n = \frac{a(n+1)}{en a(n)}, \quad C_n = \frac{a(n)}{n^{n-2}}.$$

[Write note, batch 7301.] Table 2 is not produced by the shipped verifier, and no program that

Table 2: Numerical orientation for the unresolved ratio problem.

n	R_n	C_n
20	0.930237015501	2.532871208108
50	0.970851708171	2.548375805426
100	0.985262506048	2.558520181274
150	0.990139152504	2.566400160982
200	0.992589718536	2.572452357804
250	0.994064010570	2.577220409158
299	0.995032124887	2.581012234240

produced it was delivered; it is numerical orientation only. The intake recomputed the row $n = 20$ exactly and agrees to all printed digits; the other rows were not reproduced.

The ratio data strongly support the OEIS conjecture, but the slow drift of C_n does not distinguish a finite limiting constant from a slowly varying factor. This motivates the following deliberately cautious refinement.

Conjecture 11.1 (Refined diagonal asymptotic). *There is a positive slowly varying function $L(n)$ such that*

$$a(n) = n^{n-2} L(n), \quad \frac{L(n+1)}{L(n)} \longrightarrow 1.$$

A stronger possibility is that $L(n)$ converges to a finite positive constant. Either statement implies the ratio conjecture after a standard expansion.

The exponent $n - 2$ is also suggested by parabolic iteration theory. A proof will likely require a uniform saddle-point analysis of the iteration matrix, not only a fixed-order formal conjugacy.

12 Formalization route

The proof is well suited to a short Lean development over truncated polynomials or finitely supported series. A suggested dependency order is:

1. Define binary Hamming weight, bitwise disjointness, and the canonical lift from $\mathbb{F}_2$ coefficients to $\mathbb{Z}/4\mathbb{Z}$.

2. Prove the multinomial valuation lemma for exponent 2^k . In a first formalization, this can be replaced by an induction on repeated squaring modulo 4, avoiding a general multinomial valuation library.
3. Define the carry transform $\mathcal{C}$ on finite truncations and prove Theorem 4.2.
4. Prove $(1 + T)^{-m}$ coefficient extraction and discharge the parity condition with the existing Lucas-theorem infrastructure, or with a direct bit-carry lemma.
5. Prove Theorem 5.2 and then the three diagonal cases.
6. Define R_s as a finite set and prove equivalence of (4) and (24).
7. Formalize the counting theorem separately; it uses only elementary periodic counting and geometric-series tail bounds.

The trusted core need not include the large numerical table. The theorem is symbolic; computation is useful only as regression protection and for printing examples.

13 Further research questions and topics

Research question 13.1 (The ratio conjecture). Prove $a(n+1)/a(n) \sim en$. Determine whether $a(n)/n^{n-2}$ converges, and if so identify its limit. A useful intermediate target is a two-sided bound $n^{n-2-o(1)} \leq a(n) \leq n^{n-2+o(1)}$.

Research question 13.2 (Higher two-adic precision). Extend the carry calculus from modulo 4 to modulo 8 and 16. At modulus 8, multinomial terms of valuation two enter, so exponents of binary weight three should appear. Is there a general theorem that modulo 2^q , every coefficient at an exponent of binary weight greater than q vanishes? The balanced-partition analysis suggests such a filtration, but carries among lower layers must be controlled.

Research question 13.3 (Odd-prime analogues). For a prime p , replace F by

$$F_p(x) = \sum_{j \geq 0} x^{p^j}.$$

Develop the analogue modulo p^2 . The multinomial valuation argument should restrict surviving exponents to base- p digit sum at most two, but the carry operator has $p-1$ nontrivial digits rather than one.

Research question 13.4 (Automatic and regular structure). Study the sequence $|R_s|$ and the two-parameter predicate $d \bmod 2^s \in R_s$. Is the resulting set two-automatic in a suitable paired encoding of (s, d) ? Can κ be represented as the value of a rational or algebraic generating function, rather than only as the rapidly convergent series (28)?

Research question 13.5 (Other diagonals). Use Theorem 5.2 to classify

$$[x^n] F^{o(n+c)}(x) \pmod{4}$$

for fixed integer offsets c . The adjacent OEIS sequences A168365 and A168366 correspond to $c = -1$ and $c = 1$ [2, 3]. The general two-bit formula already solves the computational part; the goal is to simplify it to residue families and counting constants analogous to R_s and κ .

Research question 13.6 (Combinatorial model). Find a direct combinatorial interpretation of the coefficients of F^{om} . For $x + x^2$, the word model behind A122888 makes lower bounds and probabilistic questions natural. A model in which powers 2^j represent allowed arities may expose the saddle point and make the ratio conjecture more tractable.

Research question 13.7 (Distinct multinomial coefficients). A separate OEIS problem inspected during this project is A086753, the number of distinct trinomial coefficients in row n , with conjectural asymptotic $(n^2 + 6n - 12)/12$ [6]. Equality of two coefficients reduces to an interval-product identity between factorial ratios. Controlling the total number of nontrivial collisions, rather than only individual collision families, remains an attractive Diophantine-asymptotic problem.

[Write note, batch 7301.] The last question is an aside: A086753 has no connection with A168362 or with the methods of this report. The conjectural asymptotic $(n^2 + 6n - 12)/12$ in that OEIS entry is Vladimir Reshetnikov's.

14 Conclusion

The sparse modulo-four pattern in A168362 is governed by a simple mechanism: modulo 4, a 2^k -th power remembers only pure powers and balanced pairs. After iteration, this forces the second binary coefficient digit onto exponents of Hamming weight at most two. Lucas's theorem then turns the parity orbit into a no-carry set, and a finite bit count yields the complete diagonal classification.

The outcome is both exact and scalable: it proves the observed OEIS pattern, predicts every future exception, gives a logarithmic counting law with an explicit constant, and applies to every entry of the iteration array. The remaining asymptotic ratio problem is untouched by the congruence proof, but the exact arithmetic structure and the reproducible computations provide a cleaner foundation for that next attack.

A Proposed OEIS update text

The following is a concise draft, not submitted to OEIS.

Let $F(x) = \sum_{j \geq 0} x^{2^j}$ and $a(n) = [x^n]F^{\circ n}(x)$. For $s \geq 1$ put $M_s = 2^s - 1$ and define $R_s \subseteq \mathbb{Z}/2^s\mathbb{Z}$ by

$$\mathbf{1}_{d \in R_s} \equiv \mathbf{1}_{d=0} + \sum_{i=0}^s \mathbf{1}_{i \text{ OR } ((i+d) \bmod 2^s) = M_s} \pmod{2}.$$

Then $a(n)$ is even for every $n > 1$. Moreover, $a(n) \equiv 2 \pmod{4}$ exactly for $n \in \{2, 3, 4, 6\}$ and for $n = 2^r + 2^s$ with $r > s \geq 1$, $r \geq 3$, and $r - s \bmod 2^s \in R_s$; all other $n > 1$ give $a(n) \equiv 0 \pmod{4}$. The next exceptional indices after the currently listed 520 are 1028, 1032, 2056, 2064, 4100, 4104, ... If $E(X)$ counts the exceptions up to X , then $E(X) = \kappa \log_2 X + O((\log \log X)^3)$, where $\kappa = \sum_{s \geq 1} |R_s|/2^s = 2.323930059308106 \dots$

B Artifact inventory

[Write note, batch 7301.] In the collection, `article.tex` is the delivered source with prefixed labels and these notes, `article.pdf` is a build of it, and `README.md` is the collection README; the delivered PDF and README, a file list `MANIFEST.txt` and the checksum list `SHA256SUMS.txt` are not shipped. The other files below are shipped under the names shown, byte-identical to the delivery.

The accompanying archive contains:

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	manuscript source and compiled PDF
<code>code/verify_a168362.py</code>	exact standard-library verifier
<code>data/verification.json</code>	machine-readable PASS record
<code>data/verification_run.txt</code>	complete verifier stdout
<code>data/residue_sets.csv</code>	R_s for $1 \leq s \leq 64$
<code>data/exceptional_indices_to_10m.txt</code>	exceptions through 10^7
<code>oeis_update_draft.txt</code>	plain-text version of the proposed update
<code>README.md</code>	build and verification instructions

References

- [1] P. D. Hanna, *A168362: coefficient of x^n in the n -th iteration of $\sum_{k \geq 0} x^{2^k}$* , The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A168362>, accessed October 1, 2026.
- [2] P. D. Hanna, *A168365: coefficient of x^n in the $(n-1)$ -st iteration of $\sum_{k \geq 0} x^{2^k}$* , OEIS, <https://oeis.org/A168365>.
- [3] P. D. Hanna, *A168366: coefficient of x^n in the $(n+1)$ -st iteration of $\sum_{k \geq 0} x^{2^k}$* , OEIS, <https://oeis.org/A168366>.
- [4] P. D. Hanna, *A112317: coefficients of x^n in the n -th iteration of $x + x^2$* , OEIS, <https://oeis.org/A112317>.
- [5] P. D. Hanna and contributors, *A122888: coefficient triangle for iterates of $x + x^2$* , OEIS, <https://oeis.org/A122888>.
- [6] J. Perry and contributors, *A086753: number of distinct entries in a trinomial slice*, OEIS, <https://oeis.org/A086753>.
- [7] N. J. Fine, *Binomial coefficients modulo a prime*, American Mathematical Monthly **54** (1947), 589–592.
- [8] R. P. Stanley, *Enumerative Combinatorics, Volume 1*, second edition, Cambridge University Press, 2012.
- [9] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [10] V. Reshetnikov, *ProveIt: formal mathematics and reproducible research artifacts*, <https://github.com/VladimirReshetnikov/ProveIt>, repository inspected October 1, 2026.